

Hysteresis in Competitive Bicycle Pelotons

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Abstract

A peloton is a group of cyclists whose individual and collective energy expenditures are reduced when cyclists ride behind others in zones of reduced air pressure; this effect is known in cycling as ‘drafting’. Through drafting cyclists couple their energy expenditures. Coupling of cyclists’ energy expenditures when drafting is the basic peloton property from which self-organized collective behaviours emerge. Here we examine peloton hysteresis, applying the definition used in the context of vehicle traffic in which a rapid deceleration to a high density state (jam) is followed by a lag in vehicle acceleration. Applying a flow analysis of volume (number of cyclists) over time, peloton hysteresis is examined in three forms: one is similar to vehicle traffic hysteresis in which rapid decelerations and increased flow (or density) are followed by extended acceleration periods and reduced flow. In cycling this is known as the accordion effect. A second kind of hysteresis results from rapid accelerations followed by periods of decreasing speeds and decreasing flow. This form of hysteresis is essentially inverse to traffic hysteresis and the accordion effect. We show this form of hysteresis using data from a mass-start bicycle points-race. A third kind of peloton hysteresis occurs when the drafting benefit is minimized on hills and weaker cyclists lose positions in the peloton, while flow/density is retained. A computer simulation shows this hysteresis among two sets of cyclist agents, each with different output capacity and models hysteresis as a peloton transitions from flat topography to a steep incline on which drafting is negligible.

Introduction

A peloton may be defined as two or more cyclists riding in sufficiently close proximity to be located either in one of two basic positions: 1) behind cyclists in zones of reduced air pressure, referred to as ‘drafting’, or 2) in zones of highest air pressure, described here alternately as ‘riding at the front’, ‘in the wind’, or in ‘non-drafting positions’. Cyclists in drafting zones expend less energy than in front positions. These zones are located either directly behind or beside at angles to other cyclists, depending on wind direction. For large pelotons (approx. >6), a proportionately higher number of cyclists will be in drafting positions, while a lesser proportion will be in front positions. Energy

expenditure when drafting as a single is reduced by approximately 18% at 32km/hr (20mph), 27% at 40km/hr (25mph), when drafting a single rider, and, in a group of eight riders, by as much as 39% at 40km/hr in a group of eight riders; drafting is negligible at speeds lower than 16km/hr (McCole et al. 1990). At the elite level, speeds of 40 to 50km/hr on flat topography are common, and pelotons of 100 or more cyclists are common.

Coupling occurs between cyclists when one or more seek the energy-saving benefits of drafting. By taking advantage of the energy savings benefits of drafting, cyclists’ energy expenditures/power outputs are thus coupled, and by alternating peloton positions to optimize energy expenditures, cyclists can sustain higher speeds for greater durations. This effectively narrows the differences in output capacities among cyclists in a peloton. This difference-narrowing is the basis for tactics and strategy in bicycle racing as cyclists seek to exploit competitors’ limited output capacities, while expending their own most efficiently.

This analysis expands on previously identified peloton phase dynamics (Trenchard 2009) and examines oscillations in peloton flow and identifies the occurrence of hysteresis in three situations.

We develop the framework for a model of peloton hysteresis based in part upon models of vehicle traffic hysteresis. A succinct description of hysteresis in traffic flows (Kuhne and Michalopoulos 1997; Treiterer and Myers 1974) is as follows:

The dynamics of traffic flow result in the hysteresis phenomena. This consists of a generally retarded behaviour of vehicle platoons after emerging from a disturbance compared to the behavior of the same vehicles approaching the disturbance.

The three hysteresis types we identify are as follows: 1. where a peloton decelerates rapidly with a corresponding flow increase, followed by a proportionately longer acceleration and corresponding flow decrease, known in cycling parlance as the accordion effect, clearly observed in criterium races; 2. where a peloton speed accelerates rapidly with some decrease in flow, followed by a proportionately longer duration of low flow even as speeds decrease - which occurs frequently in mass start races, and which we

demonstrate with data from a mass-start bicycle race on a velodrome; 3. where a peloton transitions from a period during which drafting benefit is high to a period when drafting benefit is high but power output remains roughly constant, such as when the peloton commences climbing an incline sufficiently steep when drafting benefit is minimal – during this transition high flow is temporarily retained as cyclist begin climbing even when drafting benefit is small. This state often precedes peloton disintegration when smaller groups form consisting of riders whose average competitive fitness is narrower than for the whole peloton (Fig. 1). Here competitive fitness is a combination of physiological fitness and skill to maintain optimal positions.



Figure 1. *Sorting of main peloton into groups of riders of nearly equal fitness.*

Hysteresis in Criteriums and Points-races

Vehicle traffic behavior and hysteresis have been alternatively examined by flow-density, flow-speed and speed-density plots, known as phase diagrams which depict pairwise relations between traffic variables of flow, density and speed (Zhang and Kim 2004). Here we develop a speed-flow phase diagram. A density-speed analysis is not used here, although it is frequently used in vehicle traffic analysis (Taylor et al. 2008). In vehicle traffic analysis, density models define density as vehicles per hour per traffic lane (Taylor et al. 2008), involving a continuous passage of vehicles past designated points of measurement. Density is a spatial analysis while flow is a temporal one, and because a mass-start bicycle race involves a finite number of cyclists on a course, in this case 26 cyclists on a velodrome, flow is better suited to this analysis, although we also refer to density for descriptive purposes. We thus use 26 cyclists as a constant over the time duration between when the first rider crosses a designated point – here the start/finish line – and when the last rider crosses the start/finish line.

We apply the equation $q=v/t$ for volumetric flow (known volume) (Ganesan 2002), where q is flow; t is time taken for known volume to pass a designated point; v is the known volume – here, the number of cyclists, which was constant at 26. Physical course parameters were also constant. Note there were two cyclists who nearly lapped the

main peloton for part of the race, one cyclist who retired from the race, and four who were lapped. These factors resulted in temporarily or permanently reduced peloton volume, but we ignore these for consistency and simplicity.

Flow here is thus established simply by the time measured in seconds between the first cyclist crossing the start/finish line and the last cyclist crossing the line.

$$q = T_{\text{last}} - T_{\text{first}}$$

This is similar to a cumulative flow model (Newell 1993), in which the function guarantees the conservation of the number of vehicles. In Newell's description:

$A(x,t)$ = cumulative number of vehicles to pass some location x by time t starting from the passage of some reference vehicle.

To show flow oscillations and hysteretic delay, we use data from the 2004 Canadian Nationals senior elite men's 30km, 90-lap points-race. Points-races are mass-start races on a velodrome, or track, in which competing cyclists accumulate points according to finishing position across the line on designated laps, here every five laps. One lap is 333.3m (3 laps = 1km).

We recorded the race using video camera with time counter. In analyzing the video recording, the recording was paused as the first cyclist in the group crossed the start/finish line, and times for each recorded. Speeds in km/h were calculated for each lap based on front cyclist lap time across the start finish line, representing the speed for the group. When the group divided into smaller groups, speeds were calculated for individual groups, but the computations for these groups are not used in this analysis. Flow was calculated for each lap, as described.

The results show that the majority of higher flow periods occurred between 40 and 50km per hour, which we identify as occurring when time spreads were of 10 seconds or less (Fig 2c). We identify lower flow rates (>10s spread) to have occurred at speeds of approximately 48km/hr to 57km/hr.

Unlike traffic density correlations in which there is a clear correlation between decreasing velocity and increasing density (Helbing 2001) as traffic is introduced onto roadways, leading to jams (Zhang 2005), peloton volumes (number of cyclists) are generally constant. Observations of equivalent increases in peloton flow (or density) occur as a result of changes in riders' power output, or environmental factors such as course constraints, obstacles, or external wind direction and speed, rather than increases in volume, as occurs in vehicle traffic.

These environmental factors are more prevalent in mass-start road races than they are in velodrome races, and are particularly well-observed in races called criteriums. These are mass-start events consisting of numerous laps of roughly 1km each around city blocks with corners of 90 degrees or more (Fig 4).



(a)



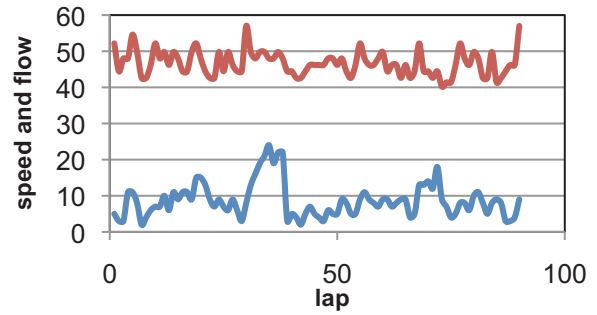
(b)



(c)

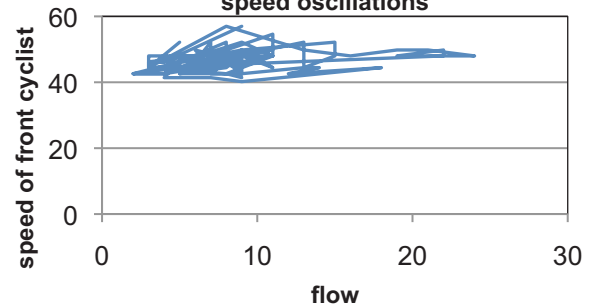
Figure 2 (a) high flow (b) medium flow (c) low flow (note two groups). Although peloton speed is slower in (a) than in (b) or (c), for the given width of the track, the peloton passed the start/finish line in approximately two seconds; approximately six seconds in (b) and ten seconds in (c) for their corresponding flow rates.

Points race - speed and flow correlation



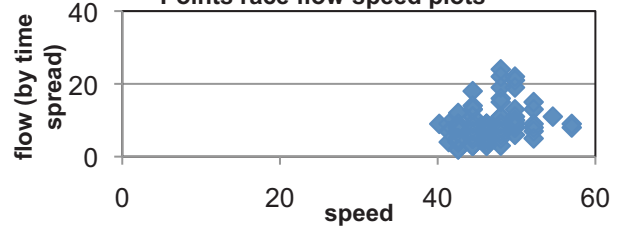
(a)

Points race - phase diagram of flow-speed oscillations



(b)

Points race flow-speed plots



(c)

Figure 3 (a) Upper curve represents speed of first rider across the line (km/h), assumed to represent the average speed of the peloton. Lower curve represents peloton flow (seconds). Together the two curves indicate a generally inverse correlation between speed and flow; i.e. flow decreases as speed increases. However, when maximal speeds were reached, as at lap 30, comparatively long periods of low peloton flow occur. In (b) speed/flow oscillations show decreasing flow following high speed periods; most prominently when speed fell from 57km/h to 48km/h, flow fell from 8 seconds to 24 seconds (note clockwise curve). In (c) flow-speed plots cluster at lower speeds and medium/high flow. Note that smaller flow values (time spread) indicate higher actual flow.

Similar to traffic jam shock-wave oscillations (Wang et al 2005; Onouchi and Nagatani 2007), shock-wave, or compression effects occur frequently in pelotons. In criteriums they occur with periodic regularity at virtually every corner for the whole duration of the race (Fig 4).

In criteriums, when riders reach a critical power output threshold, an entire peloton may form a single paceline in which riders ride one behind the other. However, speeds do not always remain sufficiently high to sustain this phase dynamic, meaning that frequently riders ride parallel or in staggered positions to each other. When cornering, generally those riders at the front of the peloton can take the optimal trajectory with minimal deceleration, while others farther back, who may approach a corner in non-optimal trajectories, must either alter their positions by temporarily decelerating to find a place in the line of riders who do take the optimal trajectory, or avoid collisions in response to other riders who are shifting their positions.

Either way, mass cyclist repositioning and collision avoidance occurs at comparatively high speed, resulting in cascading decelerations among cyclists farther back who find themselves experiencing high density, known as bunching, around corners. In turn, this deceleration requires a proportionately longer period for cyclists in the peloton to accelerate out of corners to match the speed of the few riders at the front who have travelled through at relatively constant speeds. In cycling parlance this is known as the accordion effect, and describes a self-organized hysteretic effect.

In a velodrome points-race, the banked track allows riders to compensate for lateral positional disadvantages by accelerating down the banking. Also turns are broad-sweeping, as noted (Fig 2), and so a single tangent line is not required for optimal speed and positioning around curves, as it is through corners in criterium races.

As a result, the points-race observations here and the flow rate increases that follow rapid accelerations (i.e. the bunching that occurs as riders decelerate after fatiguing from the rapid acceleration) represent a form of hysteresis that results primarily from temporary limitations in riders' competitive fitness, while the criterium accordion effect results from a combination of course constraints and limitations in rider competitive fitness, which includes the skill required of riders to find and hold optimal positions.

In the points-race data shown here, there is a gradual long-term trend toward speed reductions and increasing flow, which, through a series of flow rate and hysteretic oscillations (Fig. 3(b)), represents an attractor state. We suggest this gradual trend is due to gradually increasing rider fatigue, but further observations over different race distances are required to determine the universality of this basin of attraction and its cause.

There were at least four major hysteretic epochs, characterized by rapid accelerations to a critical threshold, followed by decreased flow and then yet further decreased flow as speeds continued to fall (Fig 3a, 3b). This dynamic

is counter-intuitive. Under normal circumstances one expects decreasing flow to correspond to increasing speed and conversely for increasing flow to accompany decreasing speed, and both of these represent the general case in peloton dynamics. But the dynamic here occurred when flow continued to decrease even as speeds decreased. The hysteretic deceleration and decreasing flow is a direct result of cyclist fatigue at a critical threshold power output.

This form of peloton hysteresis is thus the inverse of traffic hysteresis in which rapid high density states (jams) are followed by acceleration lags and decreasing density. Vehicle traffic hysteresis is illustrated by a periodic orbit loop structure, or a triangular trajectory on a speed-density plot, described as the fundamental diagram (Zhang et al. 2005).

As between a peloton and vehicle traffic, one distinguishing feature of the peloton is its competitive nature in which riders are simultaneously motivated to decrease density, for those ahead, and to increase or to maintain high density, for those behind. Cyclists also fatigue and exhibit competitive fitness thresholds when differences in fitness levels are equalized through coupling and the benefits of drafting. Vehicle traffic has effectively unlimited energy supply and there is no significant incentive to draft, and close proximity driving is both dangerous and undesirable.

In the points-race examined here, the first significant hysteretic delay occurred after rapid accelerations, most notably from 43km/h to 57km/h. In Figure 3b the curve is clockwise, and shows that the peloton flow rate generally decreased as speed increased (note decreasing flow means increased time spread between first and last riders). The first major lag began when cyclists achieved maximum speed (57km/h) and flow rate was eight seconds (note: supplementary data available). Tracing the curve from the highest point (57km/hr) down and to the right, between laps 31 and 38, peloton flow fell from 8 seconds to a maximum of 24 seconds as speed fluctuated between 48 km/h and ≈ 50 km/hr. During this time the peloton had divided into as many as nine groups. The peloton maintained this threshold speed/maximum flow for approximately eight laps (Fig 3a). Of the four major hysteretic epochs, this was the most significant as flow continued to fall for this extended period even while cyclists decelerated.

Not until peloton speeds fell to ≈ 43 km/h during laps 41-42 did groups reintegrate, when flow increased maximally to 2 seconds. Then between laps 43 and 54 flow was relatively stable, fluctuating between 5 and 9 seconds, while speeds fluctuated between ≈ 44 km/h and 48km/h. At laps 55-56 a second significant hysteretic delay occurred when speed increased to ≈ 52 km/h while flow was 9 seconds, and then decreased to 11 seconds when speed dropped to 48km/hr on lap 56. A third similar delay occurred between laps 68 and 70, when flow increased to 13 seconds at ≈ 52 km/h and then decreased to 14 seconds as speed fell to 44km/hr.

Generally, periods of low flow and comparatively high speed result from the inherently competitive nature of the event: once peloton divisions have occurred, groups in front are motivated to remain in front, while groups behind are motivated to catch those ahead. Then, at a threshold level of cyclist fatigue and reduced speed, groups reintegrate and flow increases. Comparatively long periods of higher flow thus result from a combination of fatigue and riders' competing density objectives.



Figure 4 *Criterium. Riders adjust positions to take single tangent through corner. If riders approach the corner along a less optimal tangent, they either decelerate to adjust their tangent or lose positions as they go through the corner due to additional distance travelled.*

Hysteresis in Transition from Drafting to Non-drafting

For a peloton in a velodrome or on a level road surface, drafting is always possible at higher power outputs. This is not so on steep inclines. As noted, energy saved by drafting is negligible at speeds of less than 16km/h, and on sufficiently steep inclines speeds are thus reduced. The precise gradients at which these slower speeds occur depend on the sustainable power outputs of the cyclists (Swain 1998), and here it is assumed that in a competitive situation riders will climb at or near maximum sustainable outputs.

On a hilly course in which riders approach a hill from a level part of the course, when riders proceed up an incline of sufficient gradient and length for momentum to be lost and for riders to experience a corresponding reduction in speed, drafting is effectively negated. For illustrative purposes, we describe drafting in this case as having been "turned-off" or "disabled". As such the relative climbing abilities of each cyclist are not equalized by drafting, and the peloton will begin to sort itself primarily according to climbing capacity and less so according to overall competitive fitness, which includes a skill component, as noted.

As they approach a hill on a race course, riders in a peloton will be generally distributed unevenly according to competitive fitness, such that not all weak climbers commence riding a hill at the rear of the peloton, and con-

versely not all strong climbers start the climb at the front. In fact weak climbers are commonly coached to begin a climb as near to the front as possible. In this way a weak climber, depending on the length and gradient of the hill, may remain in contact with the peloton as it crests the hill, even if he or she loses considerable ground within the peloton in the process.

The process of weaker riders slowing on a hill while remaining part of the peloton system is a hysteretic one. Riders within the peloton change position, but the average flow or density of the peloton does not change. Thus there is a lag time between the moment when drafting is minimized but power output remains high (i.e. when riders begin climbing at speeds of <16km/hr) and when drafting benefit increases significantly.

The Simulation

Using a simple computer model, we tested the hysteretic effect that occurs when cyclists approach a hill and begin climbing with minimal drafting benefit.

In this model one agent (blue) was programmed to move faster than another agent (green). Blue was programmed to move randomly either two spaces two out of three moves, or three spaces, one of three moves. Green moved two spaces only unless drafting. We refer to these moves as the agents' basic moves. Moves were thus 6 green for every 7 blue (6/7), meaning green's proportionate speed was 85.7 percent of blue's speed.

Firstly, to establish this differential in speeds between agent types, we tested this over 100 moves (Fig. 5)

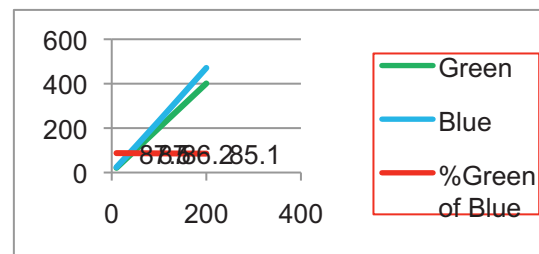
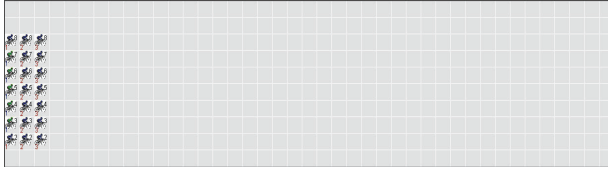


Figure 5 *Blue cyclists indicated by upper curve. Green cyclists indicated by middle curve. Predicted speed of green to blue is 6/7, or 85.7 %, based on this: for every three moves, blue moves 7 steps, and green moves 6 steps. Graph shows speeds settle near the predicted value over 200 steps. Lower curve (horizontal line) indicates proportionate speed of green to blue.*

Next we conducted three types of tests.

Test 1 Simulate drafting. We programmed green to match the speed of blue and to move in the direction of blue and match blue's moves in three positional situations: either one space directly behind blue, or at diagonals to blue, or two spaces directly behind or at one diagonal and two spaces behind blue in either diagonal direction behind blue. We refer to these as the drafting rules.

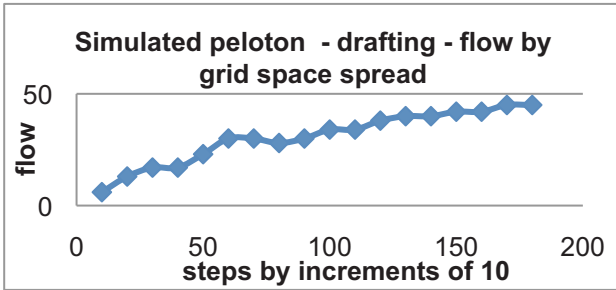
Blue moved randomly in any direction ahead according to its basic rules, which means that green agents were not always in positions that matched the speed of blue agents. To demonstrate clearly that speed matching (drafting) occurred according to the algorithm, we used five green and 16 blue. We selected these quantities to ensure that green had maximal positional opportunity to draft when drafting rules were engaged. The start positions are shown in Figure 6a. This start position was used for the three tests.



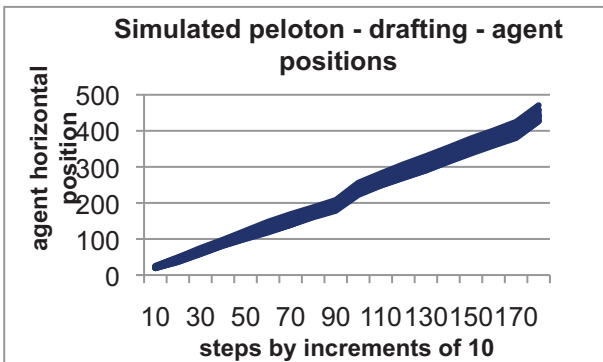
(a)



(b)



(c)

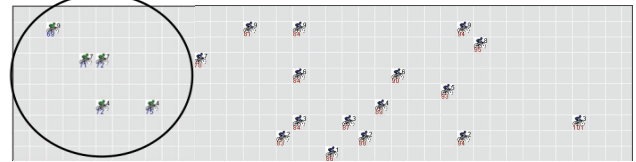


(d)

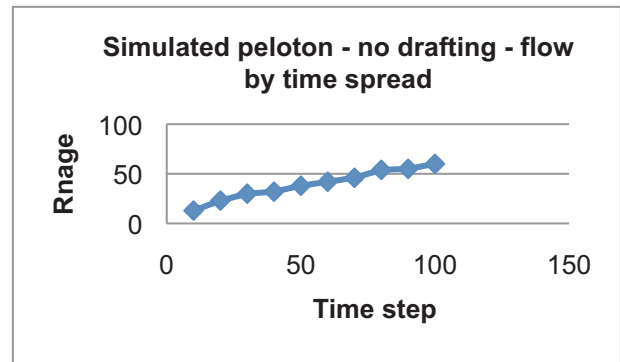
Figure 6(a) Start position for three tests. Five green drafting agents started in the far left column with one blue above and one

below. (b) After 180 steps, green agents remained mixed within the group (encircled). This indicates the drafting effect was well engaged. (c) Shows the comparatively gradual increase in grid spread. (d) Trajectories of blue and green agents. Invisible green trajectory curves are mixed with blue curves.

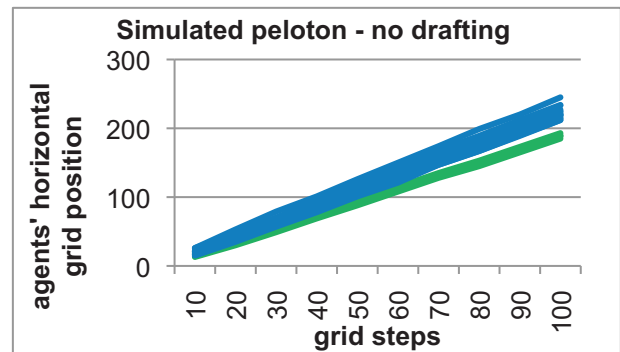
Test 2 Simulate effects of no drafting between stronger and weaker cyclists. Here the drafting rules were disabled for the duration of the test, and the simulation was run with each of blue and green proceeding according to their basic rules.



(a)



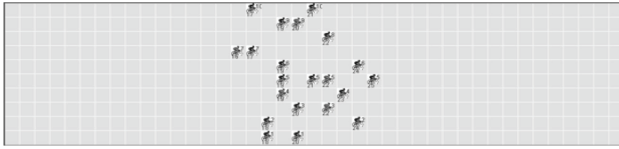
(b)



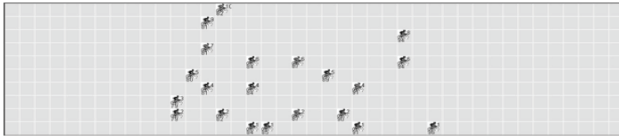
(c)

Figure 7 No drafting. In (a) after only 40 steps, green sorted into its own group behind blue (circled), as shown by lower set of curves in (c). In (b) the spread distribution is considerably greater than in the previous drafting demonstration: after 100 steps the spread is 60, compared with just over 40 after a longer period, 180 steps.

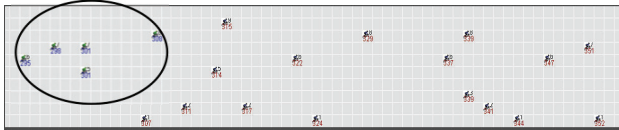
Test 3 Start with drafting, then disable drafting at step 40. Here the simulation was run with the drafting rules enabled for the first 40 steps. Then drafting rules were disabled at the completion of the 40th step.



At step 10, green agents and blue agents were well mixed.



Drafting disabled at step 40, green agents and blue agents remained well mixed, although the distribution density had decreased.



By step 150 all slower green had sorted into their own cluster (encircled).

The model shows continually decreasing flow to an equilibrium low density (not shown here) for same agent types, and infinitely decreasing flow as between stronger and weaker agents once weaker cyclists are outside drafting range of stronger cyclists.

This models the peloton dynamic when riders of different strengths travel, at one moment, at sufficiently high speeds to draft and to equalize sustainable outputs (flat topography or shallow inclines), and who then proceed up an incline sufficiently steep to minimize drafting. As riders ride up the incline, weaker riders in the midst of the group will begin to decelerate relative to others. This deceleration, however, will not immediately affect peloton density until these riders fall outside the potential drafting range of faster riders. The density or flow of the peloton is thus retained for some transitional period and actual peloton data is predicted to show a drop in flow and distribution or sorting according to competitive fitness. Clustering will occur, as in Figure 1.

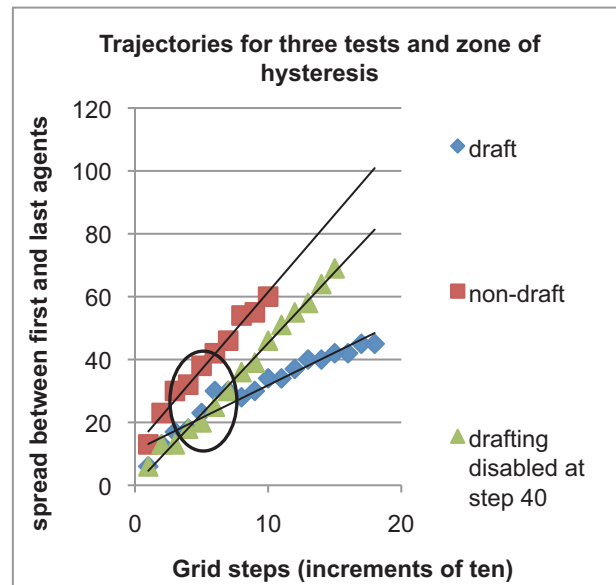


Figure 8 Circle indicates region of hysteresis, between steps 40 and 70. Here upper curve (squares) represents plots for test procedures in which drafting was disabled from the start. Lower curve (diamonds) represents plots for test procedures in which drafting was enabled for green agents from the start. Middle curve (triangles) represents plots for test procedures in which drafting was enabled for green agents from the start, but was disabled at after step 40 was complete.

Conclusion

We have identified three kinds of hysteresis effects in bicycle pelotons. The first is observed most easily in criteriums and occurs predictably and periodically, primarily as riders enter and exit corners. This dynamic, known as the accordion effect in cycling, most closely resembles vehicle traffic hysteresis in which a lag in acceleration and decreasing density follows a rapid increase in density (jam).

The second kind of hysteresis is observed unpredictably and aperiodically in all mass-start bicycle races, although we have isolated the effect in the context of a velodrome points-race. It is characterized by a delay in the reintegration or increase in flow (increase in density) as riders decelerate after a rapid acceleration. It is essentially the inverse process of vehicle traffic hysteresis.

A third kind of hysteresis is observed when riders proceed up sufficiently steep hills, and occurs as a peloton transitions from a globally coupled state - coupled by the drafting benefit - to a state in which drafting is minimal. For a time during this transition, the peloton retains a specific flow rate or density as weaker riders effectively shift positions backward in the peloton, after which the peloton divides into groups.

The focus of our analysis has been on the second and third forms of hysteresis, as we have not presented data for the criterium accordion effect.

By using data from a points-race we have shown a form of hysteresis where rapid acceleration and decreasing flow are followed by proportionately longer deceleration periods during which comparatively low flow states were retained.

Next, a computer simulation demonstrated hysteresis among two sets of cyclist agents, each with markedly different maximum fitness capacities. These were tested first while drafting, and then with no drafting, and then as drafting was eliminated part way through the test run. The last of these tests modeled the situation when a peloton transitions from flat topography to a steep incline on which drafting is negligible.

We may conclude that the first kind of hysteresis results more from peloton spatial adjustments due to course constraints, than it does from intrinsic limitations in cyclists' competitive fitness. Course parameters being a major determinant of this type of hysteresis, its analog to traffic hysteresis is more obvious, as vehicle traffic constraints are largely externally determined, as traffic systems are non-competitive and non-fatiguing with effectively unlimited energy supply with no significant incentive to draft, and high density is an undesirable state.

In contrast, the second of these hysteretic processes is driven largely by limitations in cyclists' competitive fitness and their simultaneously opposing objectives to maintain density, for those behind, and to decrease it, for those ahead. The third results from intrinsic differences in physiological fitness which are exposed in situations when the equalizing effects of drafting are minimized.

In all cases, having identified the basic parameters for the observed effects, further data is required for a more complete understanding of the dynamics and further work may be done to model the effects mathematically.

In general, we conclude that peloton hysteresis is a self-organizing dynamical process within competitive systems in which energetic coupling occurs. It occurs in different forms, and its oscillating recurrence indicates the property is resilient and robust. We may predict these kinds of hysteresis to be observable in rapidly moving herds, flocks, and sperm aggregates, among other biological collectives.

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Figure 1 Graham Watson, with permission.

Figure 2(a-c) 2004 Track Nationals, Victoria BC, Trenchard, H.

Figure 4 March 2009 Criterium Albany, Georgia. Trenchard H.